



## Evaluation of biofouling behavior of zwitterionic silane coated reverse osmosis membranes fouled by marine bacteria

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### ABSTRACT

Zwitterionic modification of membrane surfaces is a promising approach to decrease membrane biofouling. In this study, biofouling properties of two commercial polyamide (PA) reverse osmosis (RO) desalination membranes coated with zwitterionic sulfobetaine silane compounds including 4-(diethyl(3-(trimethoxysilyl)propyl)ammonia)butane-1-sulfonate (EPBS) and 3-(dimethyl(3-(trimethoxysilyl)propyl)ammonia)propane-1-sulfonate (MPPS) were investigated using five different marine bacteria isolated from Bosphorus, Turkey. To mimic the real environment, new marine strains were isolated for the biofouling assay. The biofouling behavior was monitored at different maturation stages of biofouling bacteria on the surfaces of the membranes. In addition, the permeability variation of the coated membranes was analyzed by measuring the flux values before and after the biofouling process compared to the uncoated, control membrane. According to the results, while commercial control RO membranes were highly prone to biofouling, zwitterionic sulfobetaine silane coated membranes showed a strong anti-bacterial effect against the isolated marine biofouling bacteria with significant bio-film adhesion resistance. When the flux value of commercial uncoated control membrane decreased by 38% after biofouling, the flux reduction was 16% and 20% in the EPBS and MPPS coated membranes, respectively.

### 1. Introduction

Biofouling, called “Achilles heels of the membranes” [1], is the most common challenge encountered in desalination plants using membrane technology. The membrane surface property, microorganism characteristics, and feed water composition have been acknowledged as key parameters affecting the membrane biofouling [2]. It is well established that the biofouling occurs through different mechanisms depending on the attraction of bacteria to the membrane surface by electrostatic interaction, hydrophobic interaction, hydrogen bonding, lewis acid-base interactions, and van der Waals interactions [2,3]. Furthermore, bacterial appendages such as Pili and Fimbriae can act as bridging agents [4]. Adhered bacteria negatively affects the salt rejection and water permeability performance of membranes by segregating sticky exopolymeric substances (EPS) on the surface [5]. Different modification techniques have been reported in the literature to overcome the

biofouling on reverse osmosis desalination membrane surfaces by introducing ‘non-adhesive’ or ‘anti-bacterial’ properties. Various monomers or nanoparticles with hydrophilic or anti-bacterial properties have been utilized to modify the surface of polyamide (PA) thin film composite (TFC) RO membranes by interfacial polymerization [6], grafting [7,8] or coating approaches [9]. In recent years, the use of polyvinyl alcohol (PVA) [9–11] and polyethylene glycol (PEG) [12,13] as additives or constituents has been reported to overcome fouling problem by sustaining a hydration barrier layer on the membrane surface. The modification of membrane surfaces with zwitterionic groups is more preferable due to higher antifouling activity than non-ionic, hydrophilic groups [14,15]. The formation of the hydration layer by electrostatic forces rather than hydrogen bonding renders zwitterionic groups superior to hydrophilic polymers in surface modification for antifouling purposes [16]. Different zwitterionic groups were used to modify the membrane surface for mitigating biofilm formation. Commercial

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aromatic PA-TFC RO membranes were modified using zwitterionic amino acid L-cysteine [16]. Li et al. covalently grafted poly-(sulfobetaine) zwitterionic groups on commercial RO membrane. They observed significant biofilm formation of *E. coli* on the initial commercial RO membrane. However, on the zwitterion-coated RO membrane, no biofilm formed [17]. Zwitterionic amino acid L-DOPA was coated on the surface of PA RO membranes and increased fouling resistance was observed [18]. In another work, Khan et al. synthesized the copolymer of poly(4-vinylpyridine-co-ethylene glycol diacrylate) and deposited it as a film on the RO membranes to reduce membrane fouling. The modified membrane demonstrated higher anti-biofouling performance and salt rejection [19]. Recently, we have reported the introduction of zwitterionic groups to commercially available RO membranes by the coating of the membrane surfaces with zwitterionic sulfobetaine silane compounds for durable anti-fouling membrane surfaces with easy cleanability [20].

Mostly, gram-negative *Escherichia coli* is used as the model bacteria to study the biofouling behavior of various membranes including seawater desalination RO membranes [17]. However, nowadays there are some attempts to create or select models of representative bacteria for biofouling studies [21]. In this study, in order to create a more realistic environment compared to *E. coli* used for biofouling tests, five species have been selected which are most attached to the membrane surface in the model environment with real sea water. Biofouling behavior of commercial PA-TFC RO membranes which surfaces modified with two different types of zwitterionic sulfobetaine silane compounds [20] was investigated with newly isolated marine bacteria for a better understanding.

## 2. Materials and method

### 2.1. Modification of commercial reverse osmosis membranes

SWC5 40 membranes from the Hydranautics company were selected as commercial desalination membranes for the membrane coating procedure. Membrane sheets composed of a support and active layer were cut from 4040 sized spiral wound modules and washed with 0.5% sulfuric acid and 1% NaOH solution respectively in order to clean membrane surfaces before coating procedure. The membranes were immersed in 5% (w/v) aqueous solution of propylene glycol for 30 min and then the excess solution was removed and the membranes were dried in an oven at 120 °C for 15 min. Then, the membrane surfaces were contacted with an aqueous solution containing 0.01 M potassium metabisulfite and 0.01 M potassium persulfate for 30 min and rinsed with deionized water [22,23]. EPBS and MPPS were sprayed onto the treated membrane surface at a concentration of 2% (w/v). Excess silane solutions were squeezed out with a rubber roller. Coated membranes were cured at 120 °C for 30 min. The schematic view of coating materials EPBS and MPPS was given in Fig. 1.

### 2.2. Isolation of biofouling marine bacteria

To obtain marine biofouling bacteria, commercial PA-TFC RO membrane (SWC5, Hydranautics) was exposed to marine water (Bosphorus) as a growth medium. Two different growth media, nutritional value-rich and -poor, were used to understand the biofilm formation more clearly. Most biofouling marine bacteria could be determined by the biofilm formed in its growth medium. Firstly, marine broth (MB) (Difco™ 2216) was dissolved in seawater (37.4 g/L) to provide a nutrient condition for marine bacteria. Secondly, the seawater was used without any additional nutrients. Membranes were soaked in both growth media, mixed at 185 rpm and incubated at 20 °C for 14 days. To ensure continuity of the nutrient marine condition, fresh seawater with the same concentration of the marine nutrient broth was prepared in a period of 3 days. After an incubation period of 14 days, membranes were taken out and the biofilm layer was obtained using

backwashing and scratching and then suspending the biomass in 10 mL Ringer solution (NaCl, 2.25 g/L; CaCl<sub>2</sub>, 0.12 g/L; KCl, 0.105 g/L; NaHCO<sub>3</sub>, 0.05 g/L) for dilution. The samples were serially diluted from 10 mL to 10<sup>3</sup> mL with Ringer solution [24]. From each diluted solution, 100 µL aliquots were spread on marine agar 2216 (Difco) and incubated at 20 °C for 3 days. After cultivation, bacterial colonies on the agar plates were examined in terms of colony color, size, shape, and topography of the surface. Different colonies were picked and spread several times to new agar plates until pure cultures of the bacterial strains were obtained. Isolates were stored in glycerol stock at –80 °C.

### 2.3. Growth kinetics of the isolates

The growth rate of isolated strains was examined in autoclaved marine media. The strains were incubated at 20 °C for 19 h while shaking at 180 rpm. The growth curves of the strains were plotted by measuring the optical density at a wavelength of 660 nm (OD<sub>660</sub>) at certain intervals in 19 h. The specific growth rates of the isolated strains during the exponential growth phase were determined by the growth curves [25].

### 2.4. Bacteria identification using 16rRNA gene sequencing

Isolated bacteria were identified by amplifying and partial sequencing of the 16S rRNA gene. The universal primers 27F(5'-AGAGTTTG ATCCTGGCTCAG-39) and 1492R(5'-GGTACCTTGTTACGACTT-39) were used [26]. The polymerase chain reaction (PCR) conditions were as follows: 94 °C for 5 min, 94 °C for 30 s, 45 °C for 30 s, and 72 °C for 60 s. The amplified products sequenced with an ABI 3100 Genetic Analyzer (Applied Biosystems, USA). Gene sequences were compared with GenBank databases using the BLASTN program. The neighbor-joining method with CLUSTAL W and MEGA Version 5 was used to construct a phylogenetic tree [27]. The 16S rRNA nucleotide sequences were deposited in the GenBank database under accession numbers KX650876-KX650880.

### 2.5. Quantification of biofouling and its effect on water flux

Biofouling experiments were carried out to analyze the propensity of membranes towards the attachment of bacteria. The growth medium was prepared by dissolving 37.4 g marine broth (MB) (Difco™ 2216) in DI water and autoclaved at 105 °C for 15 min. Isolated marine bacteria were grown simultaneously at 30 °C while shaking (300 rpm). Optical density was monitored and the bacteria were maintained in the growth phase by adding fresh aqueous marine broth periodically (Fig. 2). Different pieces of uncoated commercial PA-TFC RO membrane (Hydranautics) and same membranes coated with zwitterionic sulfobetaine functional trimethoxysilane compounds including EPBS and MPPS, which was reported in our previous study in detail [20], were cut and then sterilized by alcohol. As shown in Fig. 2, pieces of each membrane were immersed in the bacterial solution when the bacteria were in the growth phase. Samples for SEM analysis, CLSM analysis, and permeability test were taken after three days and ten days of biofouling.

To determine the extent of bacterial adhesion and to evaluate the biofilm formation on membrane surfaces, DI water permeability values before and after the biofouling experiment were compared. DI water flux experiments were performed in a dead-end stirred cell filtration (Sterlitech, HP4750) apparatus with the effective membrane area of 14.8 cm<sup>2</sup>. Before the biofouling test, DI water permeability of membranes has been measured at 20 bar pressure, 25 °C DI water temperature, with the pH value of 6.8 while the stirring magnet was rotating at 500 rpm. 3rd and 10th days of biofouling tests, membrane coupons were taken from the bath and DI water permeabilities were measured again. Each experiment was repeated three times. The flux reduction rate was calculated from the differences between the first and last condition.

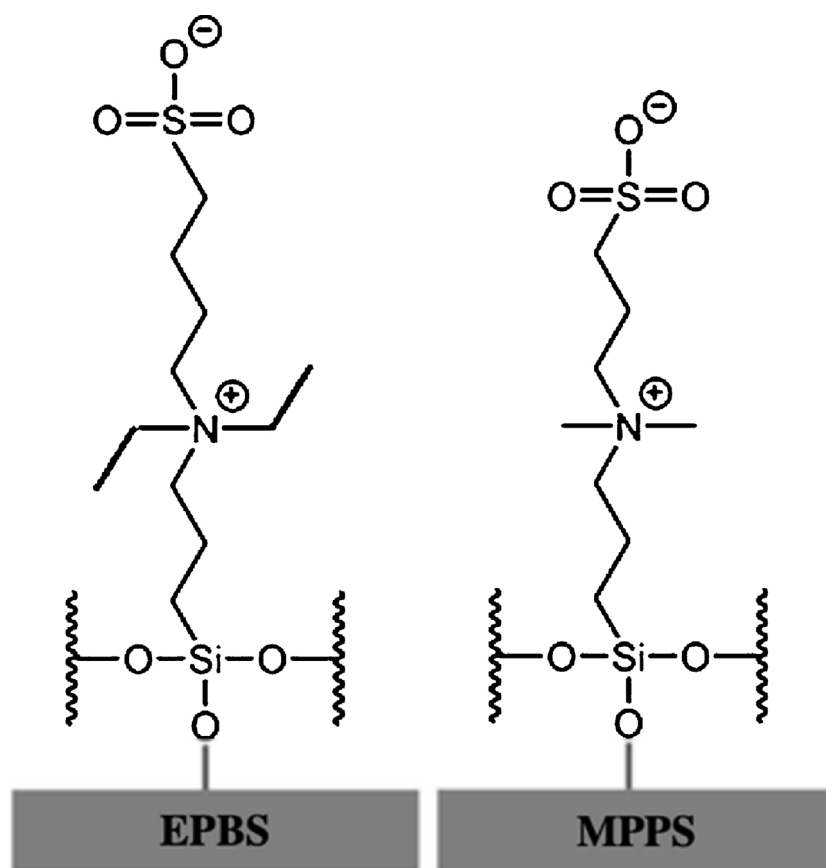


Fig. 1. Schematic view of coated materials EPBS and MPPS.

## 2.6. Visual biofilm analysis

Samples were fixed by using a glutaraldehyde solution to maintain their natural structure. Then, gradual dehydration was done by sequential immersion in various grades of aqueous acetone solution to inhibit the biofilm collapse by the high surface tension of water. After the immersion in 100% acetone, the acetone was removed from the samples at room temperature to prepare for further sputter coating. Samples were coated by sputtering gold. Images were obtained by the SEM device (Philips-XL30 SFEG).

To visualize live and dead bacteria adhered and evolved from bacteria community on membrane surfaces, fluorescent labeling technique was used with confocal laser scanning microscopy (CLSM) (Nikon C4). Samples were stained with green fluorescent nucleic acid stain Syto9 to label both live and dead bacteria and counter-stained with red

fluorescent nucleic acid stain propidium iodide (PI) to label dead bacteria (Backlight). Samples were analyzed by CLSM device equipped with Nikon Plan Apo 60<sub>1.40</sub> objective, detectors, and filter.

## 3. Results and discussion

### 3.1. Isolation and identification of marine biofouling bacteria

Five new marine biofouling bacteria obtained from the biofilm layer (M1, M2, M3, M4, M5) were isolated from the Bosphorus sea water in Istanbul, Turkey. To identify the taxonomical relationship between the sequences of 16S rRNA gene of the five new isolates A BLAST search was done. According to the results, all isolates belonged to *Gammaproteobacteria* and were classified into five genera and five species as shown in Fig. 3.

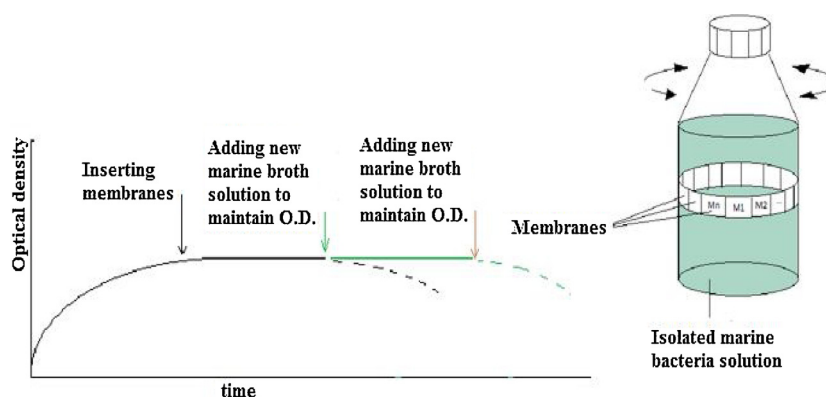
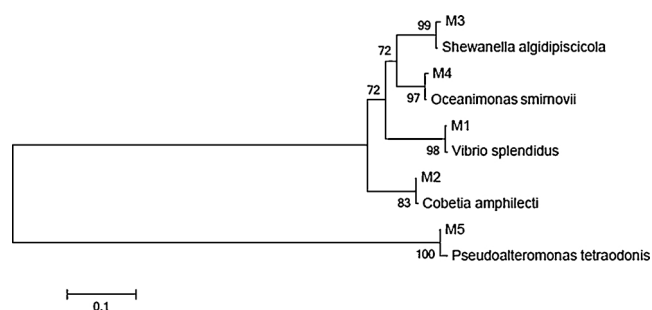


Fig. 2. Schematic illustration of the biofouling protocol.



**Fig. 3.** Neighbor-joining tree based on the 16S rRNA gene, providing phylogenetic relationship among marine bacteria that cause biofouling from Bosphorus.

**Table 1**

Accession numbers of the isolated bacteria from the biofilm layer on membrane surfaces.

| Bacteria                                | Accession numbers |
|-----------------------------------------|-------------------|
| <i>Vibrio splendidus</i> M1             | KX650876          |
| <i>Cobetia amphilecti</i> M2            | KX650877          |
| <i>Shewanella algidipiscicola</i> M3    | KX650878          |
| <i>Oceanimonas smirnovii</i> M4         | KX650879          |
| <i>Pseudoalteromonas tetraodonis</i> M5 | KX650880          |

The assessment of the partial nucleotide sequence of the amplified 16S rRNA gene revealed a strong homology of the following genera: *Shewanella*, *Vibrio*, *Oceanimonas*, *Cobetia*, *Pseudoalteromonas*. Accession numbers of the isolated strains were given in Table 1.

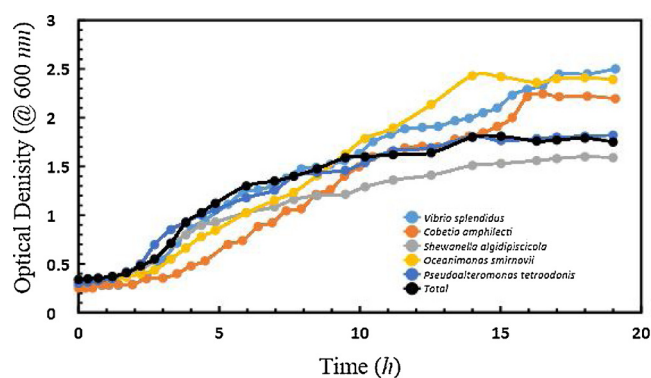
The newly isolated strains belonged to the genera that had previously been reported as a marine microorganism. The strain M1 identified as *Vibrio splendidus* is a dominant culturable *Vibrio* found in coastal marine sediments, seawater and bivalve organisms [28]. The strain M2 identified as *Cobetia amphilecti* is gram-negative, aerobic, oxidase-negative, catalase-positive, non-pigmented, rod-shape celled, 0.8–0.9 µm in diameter and 1.1–1.3 µm in length, motile by means of one polar and/or two or three lateral flagellar [29]. The strain M3 was identified as *Shewanella algidipiscicola*, which is isolated from marine fish (cod and plaice) in the Baltic [30]. The M4 strain was identified as *Oceanimonas smirnovii*, which is mesophilic bacterium of the genus *Oceanimonas* isolated from the seawater sample collected in the Karadag Natural Reserve of the Eastern Crimea, Black Sea [31] (Fig. 4).

### 3.2. Growth rates of isolated strains

Growth rates of isolated bacteria in their marine medium was determined to observe their microbial characteristics. As shown in Fig. 5, *Vibrio splendidus* M1 had the highest growth rate among all isolates. The specific growth rate constants of the strains were determined during the exponential growth phase and given in Table 2.

### 3.3. SEM analysis of membrane surfaces

Before biofouling, SEM images of virgin membranes including commercial unmodified control membrane, EPBS coated membrane



**Fig. 5.** Comparison of the growth rates of the newly isolated five marine biofouling bacteria. The black line shows incubation result of all mixed bacteria at 600 nm OD.

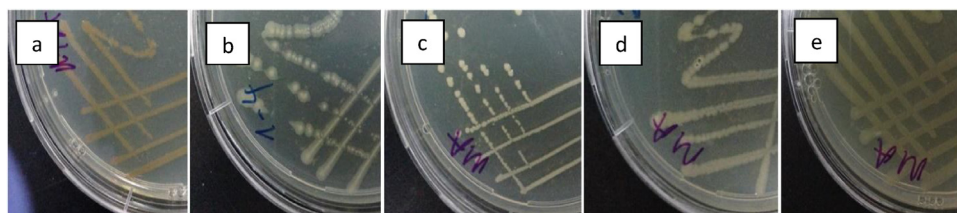
**Table 2**

The specific growth rate constants of the isolated bacteria.

| Bacteria                             | Growth rate constant |
|--------------------------------------|----------------------|
| <i>Shewanella algidipiscicola</i>    | 0,22                 |
| <i>Pseudoalteromonas tetraodonis</i> | 0,27                 |
| <i>Vibrio splendidus</i>             | 0,41                 |
| <i>Oceanimonas smirnovii</i>         | 0,41                 |
| <i>Cobetia amphilecti</i>            | 0,38                 |
| Total                                | 0,27                 |

and MPPS coated membrane are as shown in Fig. 6. For each membrane (uncoated, EPBS coated and MPPS coated), samples were taken for SEM analysis on the third and tenth day of biofouling, and their images are shown in Fig. 7 and 8, respectively.

At the end of the third day of biofouling, the bacterial adhesion on the uncoated commercial membrane surface was more pronounced than coated membranes, demonstrating the fact that zwitterionically coated membranes showed less propensity to the bacterial adhesion. As seen in both Fig. 7 and 8, 2% EPBS coated membrane had more bacterial adhesion than the 2% MPPS coated membrane. In the membrane biofouling, bacteria adhere to the membrane surface by finding a nutritionally desirable substrate to colonize. In fact, the approach of bacteria towards the membrane surface occurs via various mechanisms depending on hydrophobic interactions, electrostatic interactions, hydrogen bonding, and lewis acid-base interactions. The effect of hydrodynamic forces was eliminated in our biofouling analysis by gently mixing the bacterial solution without any pressure. SEM images obtained at the end of the third day of biofouling primarily represent the adhesion stage of the biofouling process. However, three days were not enough for bacteria to grow on the surface of membranes. The adhesion of bacteria is a critical step in the biofouling process, which triggers the further multiplication of bacteria on the membrane surface. Bacterial adhesion and the multiplication of bacteria on the membrane surface are highly affected by the surface characteristics of membranes such as surface roughness, hydrophilicity, and charge. It is important to note that when the biofouling was performed without exerting any cross-flow filtration and pressure, the effect of surface roughness should not be taken into consideration [20].



**Fig. 4.** Colonies of isolated bacteria from the biofilm layer on the seawater RO membrane after biofouling. (a) *Vibrio splendidus* M1 (b) *Cobetia amphilecti* M2 (c) *Shewanella algidipiscicola* M3 (d) *Oceanimonas smirnovii* M4 (e) *Pseudoalteromonas tetraodonis* M5.

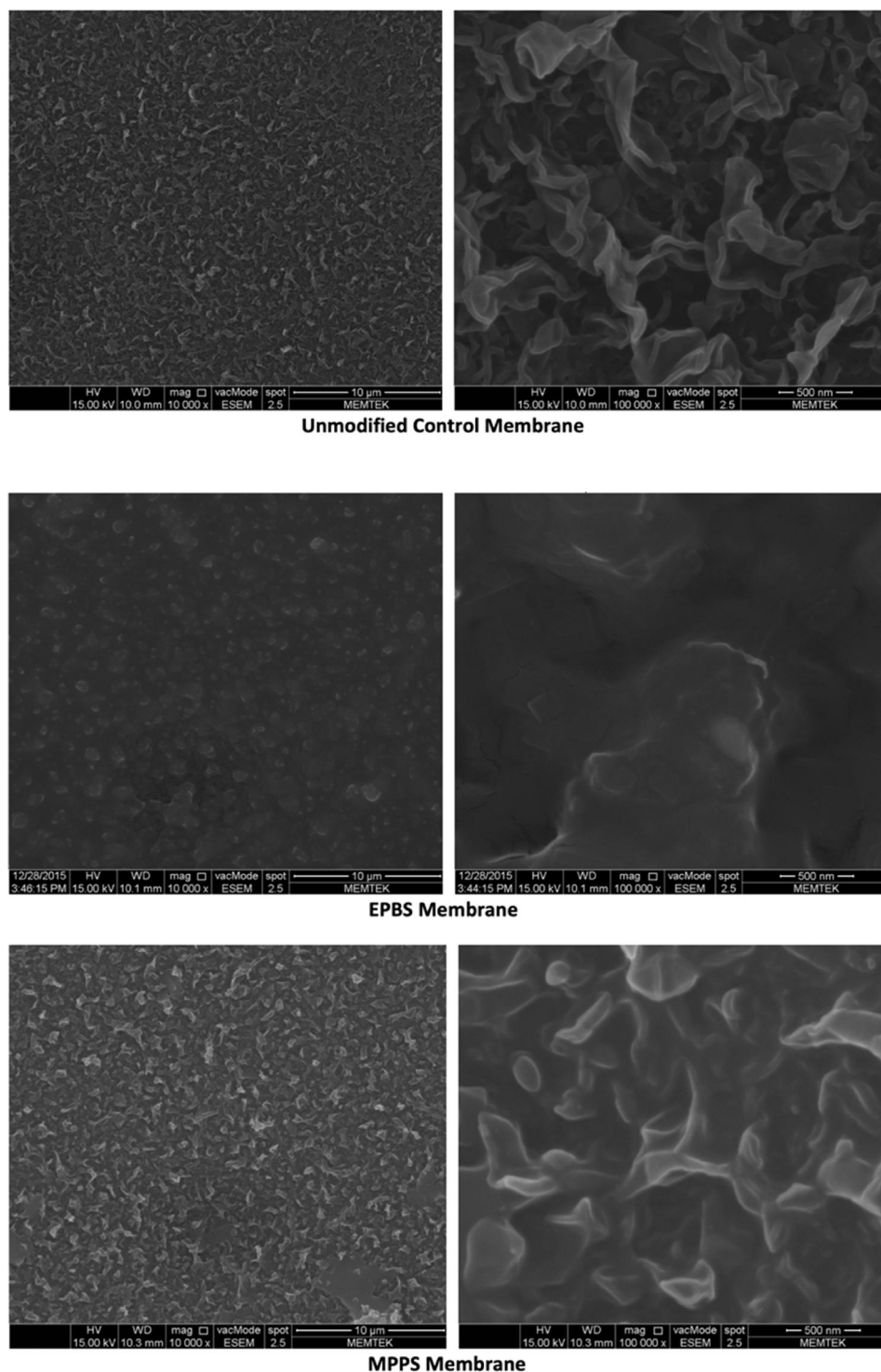
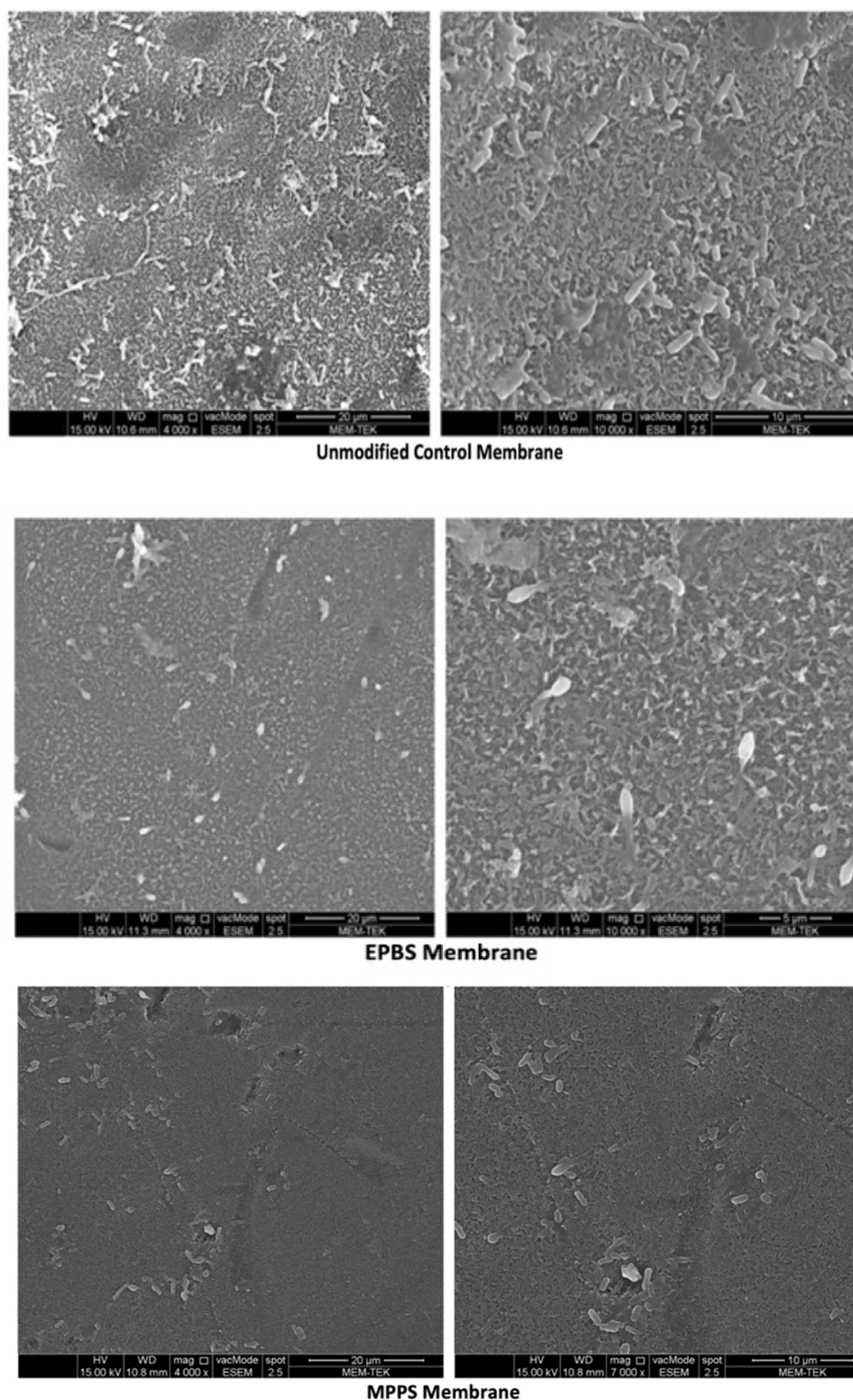


Fig. 6. Before biofouling, SEM images of virgin membranes magnified at 10,000 $\times$  and 100,000 $\times$ .

The formation of the hydration layer on the zwitterionic surface functional membranes creates an energy barrier [32,33] and holds bacteria and organics away from approaching towards the membrane surface as illustrated in Fig. 9. The effect of the zwitterionic modification on biofouling is considerably reflected in the SEM analysis of the membranes after the tenth day of biofouling shown in Fig. 8. The commercial membrane surface was densely covered by bacterial aggregates while the coverage of EPBS and MPPS coated membrane surfaces was significantly lower with several isolated bacterial aggregates.

### 3.4. CLSM analysis

Different maturation stages of bacteria were observed by CLSM analysis to evaluate its ability of biofilm formation on the commercial and modified membrane surfaces. The visualization of living or dead bacteria was performed by green fluorescent staining of living and dead cells' DNA and red fluorescent counterstaining of nucleic acids of dead bacteria as illustrated in Fig. 10. Images were obtained after the third and tenth day of biofouling. As shown in Fig. 10, at the end of the third day of biofouling, the commercial membrane was fully covered by



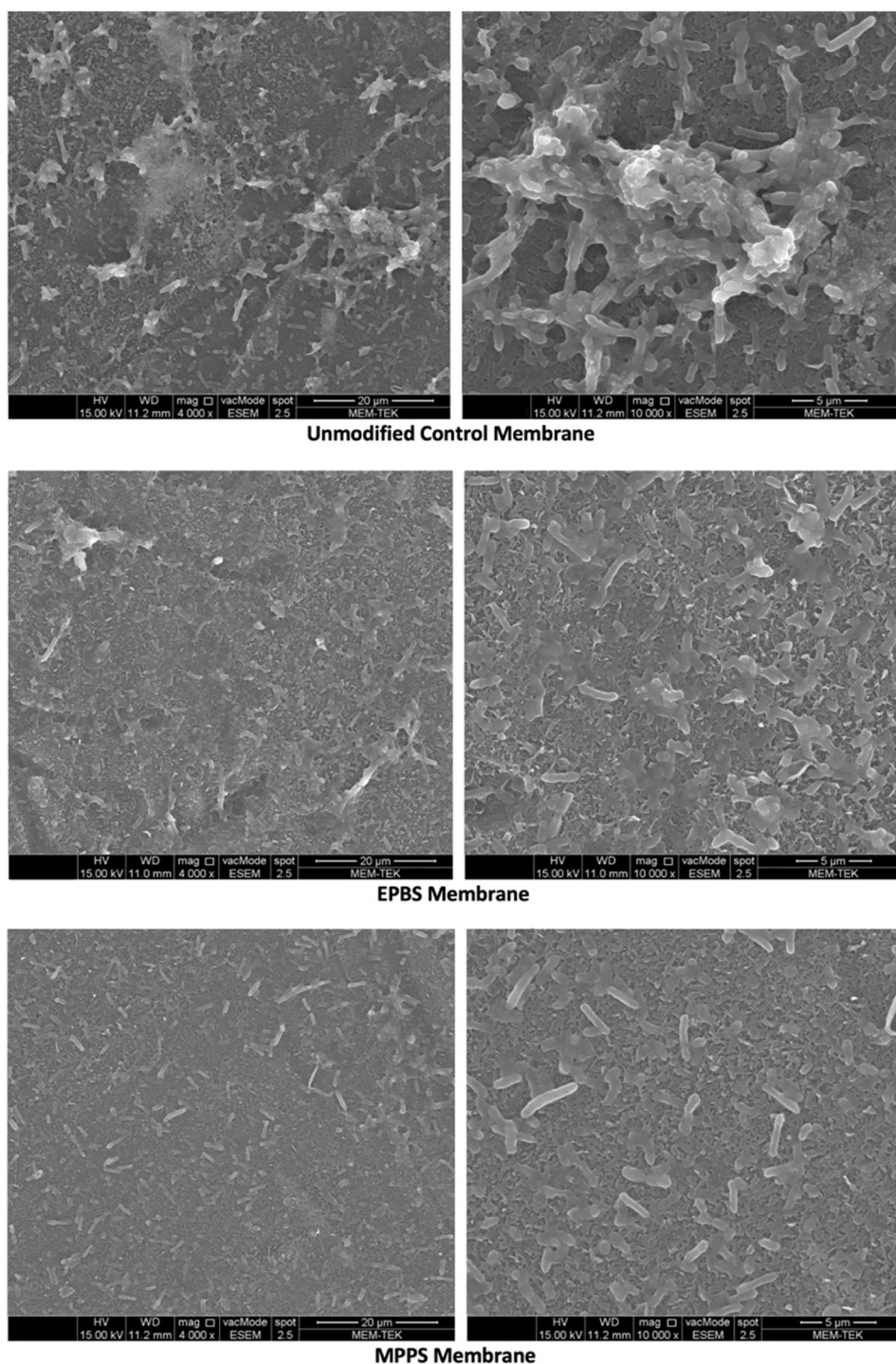
**Fig. 7.** SEM images of the unmodified membrane, EPBS coated membrane and MPPS coated membrane obtained at 3rd day of biofouling with the mixture of all five biofouling bacteria.

living bacteria. However, zwitterionic silane coated membranes exhibited increased resistance to bacterial adhesion, as observed in SEM images. The red colors of residual biomass on the EPBS-2% and MPPS-2% coated membranes show the dead bacteria. The presence of dead bacteria on these membranes was attributed to the antibacterial property of positively charged quaternary ammonium functional groups. Thus, sulfobetaine functional zwitterionic groups decreased the

biofouling not only by anti-adhesion property but also antibacterial effect as illustrated in Fig. 10.

### 3.5. Permeability variation by biofouling

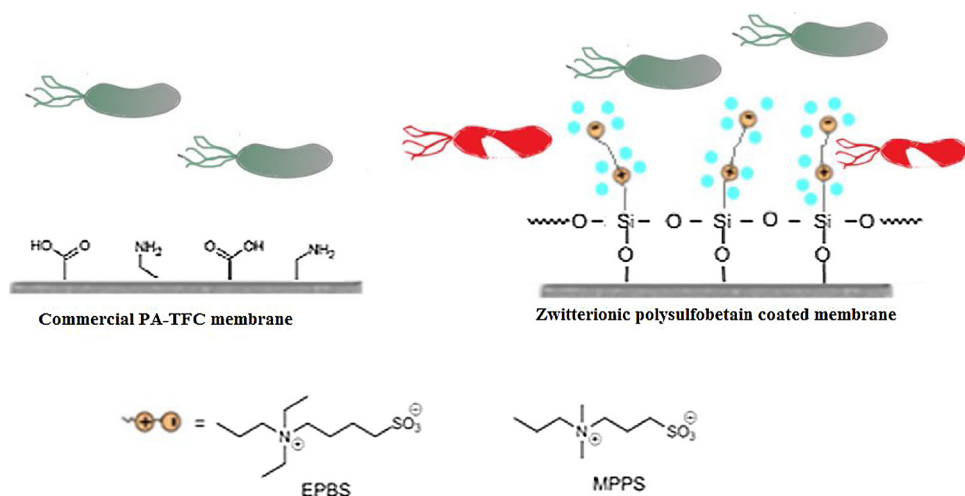
The bacterial population in the biofilm is a determining factor in the flux decline. It is well known that the bacteria in the biofilm



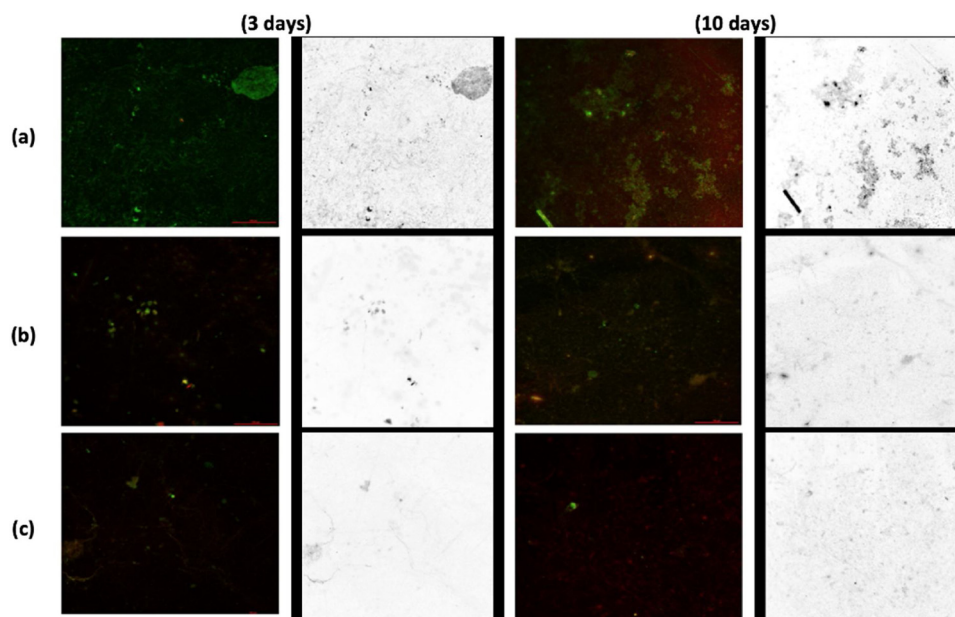
**Fig. 8.** SEM images of the unmodified membrane, EPBS coated membrane and MPPS coated membrane obtained at 10th day of biofouling with the mixture of all five biofouling bacteria.

communicate with each other by quorum sensing and start to excrete EPS rapidly when its population reaches a certain level [34]. The water flux performance of the membranes was analyzed before and after ten days of biofouling. As expected, DI water flux of all membranes at the end of this period decreased due to the development of the biofilm layer on the membrane surfaces according to Fig. 11. The flux value of commercial, uncoated membrane decreased significantly, up to 38%. However, the flux decline was 16% and 20% for EPBS and MPPS coated membranes, respectively. The results of the permeability test were in accord with the results of SEM and CLSM analysis. The main cause of flux reduction is the EPS excreted by adhered bacteria. It is believed

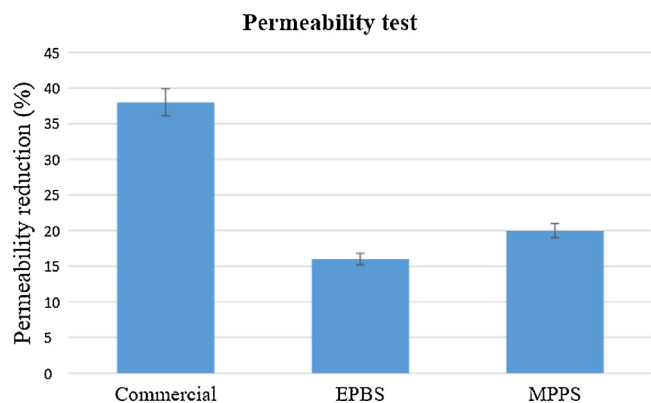
that EPS excretion varies depending on the type of bacteria. Due to the competition between bacteria, the bacterial community composition may vary at the initial stages and at the maturation stages of biofouling bacteria. Consequently, the excreted EPS and flux reduction become highly dependent on the microorganisms and the membranes' surface properties. However, due to the similar rod-shaped morphology of all the bacteria used in this study, it was not possible to trace and distinguish the bacterial species to determine a composition on different membranes with the existing visualization devices.



**Fig. 9.** Schematic illustration of the biofouling on commercial PA membrane and zwitterionic modified membranes. Red bacteria represent dead bacteria and green bacteria represent living bacteria. Blue water molecules represent the hydration layer (This figure was illustrated using scientific data given by Hadidi et al. 2014 and Ginic-Markovic et al. 2015 [32,33]) (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).



**Fig. 10.** CLSM images and inverted white and black images of (a) unmodified control membrane (b) EPBS coated membrane (c) MPPS coated membrane obtained at 3rd and 10th day of biofouling.



**Fig. 11.** DI water permeability variation of membranes after 10 days of biofouling.

#### 4. Conclusions

Biofouling properties of commercial RO membranes coated with zwitterionic silane compounds including EPBS and MPPS were investigated by SEM and CLSM analysis at the end of three and ten days of biofouling and the results were compared with the uncoated control membrane. Zwitterionic silane coated commercial RO membranes displayed higher resistance to biofouling than the uncoated, commercial control membrane. SEM analysis revealed that less bacterial adhesion and growth occurred on the zwitterionic sulfobetaine silane coated membrane surfaces. Less adhesion was attributed to the formation of a hydration layer on the coated membrane surfaces by zwitterionic groups. The results of the CLSM analysis confirmed the results of SEM analysis and also revealed the antibacterial effect of the sulfobetaine zwitterionic groups due to the presence of positively charged quaternary ammonium moieties. Finally, the permeability studies before and after biofouling with new bacteria isolated from Bosphorous showed that the anti-biofouling effect of zwitterionic modification results in a lower reduction in permeability after biofouling compared to the uncoated control membrane.

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